

VISHWA PERNAPATI

AI Engineer | .NET (C#) | Microsoft Azure | Oracle Database

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PROFESSIONAL SUMMARY

- AI Engineer with 13+ years of experience building enterprise-grade AI/ML solutions using .NET Core (C#), Microsoft Azure AI services, and Oracle Database — with deep, hands-on Oracle PL/SQL expertise spanning stored procedures, packages, triggers, views, and query tuning across large-scale enterprise systems.
- Extensive experience integrating AI/ML models with Oracle DB backends — feeding structured Oracle data into RAG pipelines, LLM inference layers, and Azure AI services for intelligent, data-driven enterprise applications.
- Proven expertise deploying and managing solutions on Microsoft Azure — Azure OpenAI, Azure AI Studio (Foundry), Azure App Services, AKS, Azure Functions, and Azure Cognitive Services.
- Strong foundation in Oracle SQL performance tuning: query optimization, execution plan analysis, index design, partitioning strategies, and stored procedure optimization for high-volume OLTP and analytical workloads.
- Skilled in building REST APIs and microservices in .NET Core (C#) that integrate seamlessly with Oracle DB and Azure AI services to deliver scalable, end-to-end AI-driven solutions.

TECHNICAL SKILLS

Oracle DB	Oracle Database 11g/12c/19c, PL/SQL, Stored Procedures, Packages, Triggers, Views, Cursors, Sequences, Oracle SQL tuning, Execution Plans, Index Design, Partitioning, Oracle Forms/Reports, SQL*Plus
AI / ML	RAG Pipelines, LLM Integration, Prompt Engineering, AI Agents, Azure OpenAI, Azure AI Studio (Foundry), Amazon Bedrock, Semantic Kernel, LangChain, AutoGen, MLflow, Copilots
.NET / Backend	C#, .NET Core / .NET 6/7/8, ASP.NET Core Web API, Python, Microservices, REST APIs, GraphQL, gRPC, Entity Framework Core, LINQ
Azure	Azure OpenAI, Azure AI Studio, Azure App Services, Azure Functions, AKS, Cognitive Services, Azure Entra ID, Key Vault, Service Bus, Azure SQL, Azure DevOps
Other Databases	SQL Server (T-SQL), PostgreSQL, MongoDB, DynamoDB, Cosmos DB, Redis, Snowflake
Cloud (AWS)	Lambda, API Gateway, ECS/EKS, S3, DynamoDB, RDS, CloudWatch, IAM, Bedrock, CodePipeline
Frontend	React 18+, Angular 10+, Next.js, TypeScript, JavaScript, HTML5, CSS3, Tailwind CSS, Bootstrap
DevOps / MLOps	Docker, Kubernetes, CI/CD, GitHub Actions, Azure DevOps, MLflow, Terraform

PROFESSIONAL EXPERIENCE

Toyota Motor North America (TMNA) | Lead Azure AI Engineer — .NET, RAG & Copilot Systems

Jun 2025 – Present | Plano, TX

Integrated Oracle Database as the primary enterprise data source for AI pipelines — writing optimized PL/SQL stored procedures and packages to extract, transform, and feed structured data into Azure OpenAI RAG pipelines and vector search indexes, enabling accurate, context-aware AI responses over live enterprise data.

Designed .NET Core (C#) microservices using Oracle Data Provider (ODP.NET) to connect AI inference layers with Oracle DB, supporting real-time data retrieval for 10,000+ AI queries per month in production.

- Architected enterprise RAG pipelines using Azure OpenAI + Azure AI Search and Amazon Bedrock + FAISS over 50,000+ internal documents; Oracle DB served as the structured data backbone for all enterprise knowledge queries.
- Built Toyota GPT-based internal copilots on React + .NET Core with Oracle-backed REST APIs, reducing analyst research time ~35% and handling 10,000+ queries/month in production.
- Implemented LangChain and Semantic Kernel orchestration layers in Python and .NET; configured Oracle DB as a tool data source for agentic workflows and autonomous task pipelines.
- Enforced responsible AI practices: prompt guardrails, content safety, bias evaluation, and Azure Entra ID / OAuth 2.0 secured APIs with Oracle-backed RBAC.
- Improved model accuracy by 22% through iterative prompt tuning, A/B experimentation, and structured model evaluation with human-in-the-loop feedback.

Pioneer Natural Resources | Lead Full Stack Developer — Azure AI & GenAI

Aug 2023 – May 2025 | Remote

Designed and optimized Oracle PL/SQL stored procedures, packages, and views to support AI-driven document search and GenAI workflows built on Azure AI Studio / Foundry — reducing manual document review time by 50%.

Built .NET Core + ASP.NET Web API services integrated with Oracle DB via ODP.NET, exposing structured enterprise data to Azure OpenAI and LangChain AI pipelines for conversational and analytical use cases.

- Wrote complex Oracle SQL queries, joins, and analytical functions (RANK, DENSE_RANK, LAG/LEAD, PARTITION BY) to prepare and aggregate datasets consumed by AI/ML models for reporting and predictive analytics.
- Designed full-stack enterprise applications using Angular + ASP.NET Core microservices and React.js + Node.js backends, deployed on Azure and AWS via automated CI/CD pipelines.
- Integrated AI copilots using Azure OpenAI and LangChain with Oracle DB-backed prompt context, ensuring enterprise compliance with safety guardrails and audit trails stored in Oracle.
- Cut deployment cycle from 2 days to under 2 hours via AWS CodePipeline and CodeBuild CI/CD automation.

Proficient INC. | Lead Full Stack Engineer — .NET, React & AI Solutions

Sep 2021 – Jun 2023 | India

- Developed and maintained Oracle PL/SQL packages, stored procedures, and triggers for healthcare and financial enterprise clients; optimized slow-running queries by analyzing execution plans and redesigning indexes, achieving up to 70% query performance improvement.

Built ASP.NET Core Web APIs backed by Oracle DB serving 100K+ daily requests — implemented Oracle connection pooling, bulk operations, and batch processing to ensure throughput and low latency.

Developed RAG-based AI search features using Pinecone and FAISS vector embeddings; Oracle DB served as the authoritative structured data source feeding the semantic search pipeline.

- Automated Oracle ETL pipelines using Python and PL/SQL scripts across Oracle and Snowflake, reducing data pipeline processing time by 60%.
- Designed AI workflows using Azure Functions, Logic Apps, and Azure ML for intelligent document processing pipelines.

Conduent | Lead Software Engineer — .NET & Cloud-Native Systems

Oct 2016 – Sep 2021 | India

Designed and managed Oracle Database schemas for enterprise healthcare and finance applications serving 200,000+ users — built complex PL/SQL stored procedures, packages, and triggers for core business logic.

Integrated Oracle DB with .NET Core and Angular applications using ODP.NET and Entity Framework; implemented caching strategies with Redis and Azure SQL to reduce Oracle load and improve API response time by 45%.

- Architected microservices using RabbitMQ and REST APIs; containerized with Docker and deployed on Azure Kubernetes Service (AKS) with Oracle DB as the central persistence layer.
- Mentored junior engineers on Oracle PL/SQL best practices, code review standards, and deployment processes across multiple release cycles.

EARLIER EXPERIENCE

Unisys — Catholic Health Initiatives | Systems Analyst (.NET & Oracle DB)

Oct 2014 – May 2016 | USA

- Developed extensive Oracle PL/SQL stored procedures, packages, cursors, and triggers for healthcare transaction processing systems, handling EDI x12 claim workflows and patient data integration across multiple hospital systems.
- Built ASP.NET MVC / Web API applications connected to Oracle Database via ODP.NET, exposing PL/SQL business logic as RESTful services consumed by Angular SPAs.
- Performed Oracle query tuning using EXPLAIN PLAN and Oracle AWR reports; redesigned key stored procedures reducing claim processing time by 40%.
- Managed Oracle schema migrations, rollback scripts, and version-controlled DDL changes across dev, QA, and production environments.

Polaris Software Labs / Citibank | Full Stack Developer (.NET & Oracle PL/SQL)

Jul 2012 – Oct 2014 | India

Core Oracle PL/SQL developer on the TRAMP transaction reconciliation platform for Citibank — designed and maintained Oracle packages, stored procedures, views, and sequences handling millions of daily financial transactions.

Implemented Oracle DB data integrity controls — constraints, triggers, and audit tables — ensuring data accuracy and compliance for financial transaction records.

- Developed ASP.NET and VB.NET application layers over Oracle DB, building batch reconciliation jobs using Oracle Scheduler and PL/SQL for automated overnight processing.
- Optimized high-volume Oracle SQL queries supporting real-time reconciliation dashboards, reducing report generation time from 15 minutes to under 2 minutes.

EDUCATION

- Master of Science, Software Engineering — San Jose State University | 2024
- Bachelor of Engineering, Computer Science — India | 2012